

Review Package Solutions

①

MOTION

1. $s(t) = -t^3 + 4t^2 - 10$

$$v(t) = -3t^2 + 8t$$

$$a(t) = -6t + 8$$

$$v(3) a(3) = (-27 + 24)(-18 + 8)$$

$$= (-3)(-10)$$

$$= 30 \Rightarrow \text{speeding up at } t = 3 \text{ s.}$$

2. $s(t) = t^3 - 6t^2 + 8t$

a) $v(t) = 3t^2 - 12t + 8$

b) $v(t) = 0$

$$3t^2 - 12t + 8 = 0$$

$$t = \frac{12 \pm \sqrt{144 - 4(3)(8)}}{6}$$

$$t = \frac{12 \pm \sqrt{48}}{6}$$

$$t = \frac{6 \pm 2\sqrt{3}}{3}, \quad t \approx 0.85 \text{ s (or)} \quad t \approx 3.15 \text{ s.}$$

(2)

2c. $a(t) = 6t - 12$
cont'd

$$a(t) = 0$$

$$6t - 12 = 0$$

$$t = 2 \text{ s.}$$

d. $a(t) = 6t - 12$

e. $\begin{array}{c} (-) \quad (+) \\ \hline 2 \end{array}$

negative: $[0, 2)$

positive: $(2, \infty)$

f. $a(t) \cdot v(t)$

$$= (6t - 12)(3t^2 - 12t + 8)$$

$\begin{array}{ccccccc} (-) & & (+) & & (-) & & (+) \\ \hline & \frac{6-2\sqrt{3}}{3} & 2 & & \frac{6+2\sqrt{3}}{3} & & \uparrow (10) \end{array}$

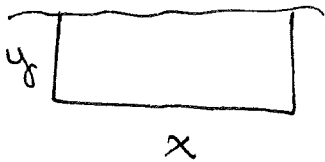
slowing down: $\left[0, \frac{6-2\sqrt{3}}{3}\right) \cup \left(2, \frac{6+2\sqrt{3}}{3}\right)$

speeding up: $\left(\frac{6-2\sqrt{3}}{3}, 2\right) \cup \left(\frac{6+2\sqrt{3}}{3}, \infty\right)$

PERIMETER & AREA

3

1.



$$x + 2y = 2400$$

$$A = xy$$

$$A = (2400 - 2y)y$$

$$A' = 0$$

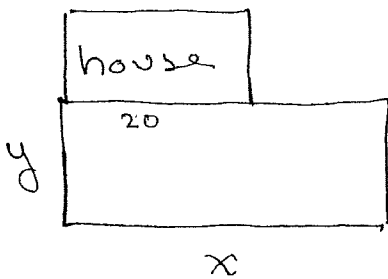
$$A' = 2400 - 4y = 0$$

$$\Rightarrow y = 600 \text{ m}$$

$$\therefore \text{Max Area} = (600)(1200)$$

$$\text{Max Area} = 720,000 \text{ m}^2$$

2.



$$xy = 600$$

$$y = \frac{600}{x}$$

$$P = 2y + x + (x - 20)$$

$$P = \frac{1200}{x} + 2x - 20$$

$$P'(x) = -\frac{1200}{x^2} + 2$$

$$P'(x) = 0$$

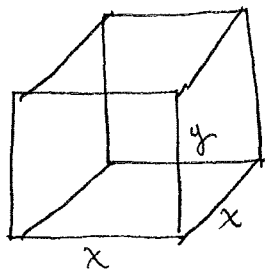
$$-\frac{1200}{x^2} + 2 = 0 \Rightarrow x = 10\sqrt{6} \text{ m}$$

$$\therefore \text{Perimeter} \approx 77.98 \text{ m}$$

Volume & S.A.

4

1.



$$V = x^2 y$$

$$4000 = x^2 y$$

$$y = \frac{4000}{x^2}$$

$$SA = x^2 + 4xy$$

$$SA = x^2 + 4x \left(\frac{4000}{x^2} \right)$$

$$SA = x^2 + \frac{16000}{x}$$

$$(SA)' = 2x - \frac{16000}{x^2}$$

$$(SA)' = 0$$

$$(SA)' = 0 = 2x - \frac{16000}{x^2}$$

$$2x^3 = 16000$$

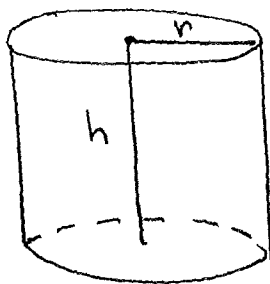
$$x^3 = 8000$$

$$x = 20 \text{ cm}$$

$$\Rightarrow y = 10 \text{ cm.}$$

(5)

2.



$$\pi r^2 h = 1$$

$$h = \frac{1}{\pi r^2}$$

$$SA = 2\pi r^2 + 2\pi r h$$

$$SA = 2\pi r^2 + 2\pi r \left(\frac{1}{\pi r^2} \right)$$

$$SA = 2\pi r^2 + \frac{2}{r}$$

$$(SA)' = 0 = 4\pi r - \frac{2}{r^2}$$

$$4\pi r^3 = 2$$

$$\pi r^3 = \frac{1}{2}$$

$$r = \frac{1}{\sqrt[3]{2\pi}}$$

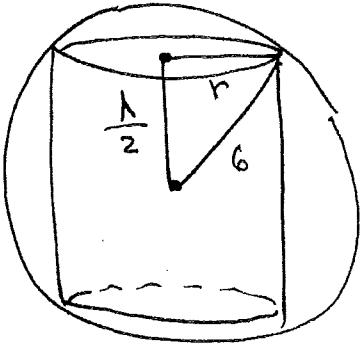
$$h = \frac{1}{\pi r^2}$$

$$h = \frac{1}{\pi \left(\frac{1}{\sqrt[3]{2\pi}} \right)^2}$$

$$h = \frac{2}{\sqrt[3]{2\pi}} \quad \therefore \frac{h}{r} = 2$$

3.

(6)



$$r^2 + \left(\frac{h}{2}\right)^2 = 36$$

$$r^2 = 36 - \frac{h^2}{4}$$

$$V = \pi r^2 h$$

$$V = \pi h \left(36 - \frac{h^2}{4}\right)$$

$$V' = \pi \left(36 - \frac{3h^2}{4}\right)$$

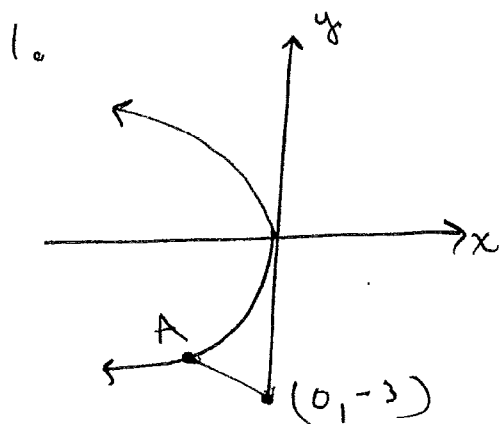
$$\pi \left(36 - \frac{3h^2}{4}\right) = 0$$

$$h = 4\sqrt{3} \text{ cm}$$

$$\therefore V = 96\sqrt{3} \pi \text{ cm}^3$$

Distances

7



$$x = -\frac{y^2}{4}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{\left(\frac{y^2}{4}\right)^2 + (-3 - y)^2}$$

$$d = \sqrt{\frac{y^4}{16} + 9 + 6y + y^2}$$

$$d' = \frac{\frac{1}{4}y^3 + 6 + 2y}{2\sqrt{\frac{y^4}{16} + 9 + 6y + y^2}} = 0$$

$$\frac{1}{4}y^3 + 6 + 2y = 0$$

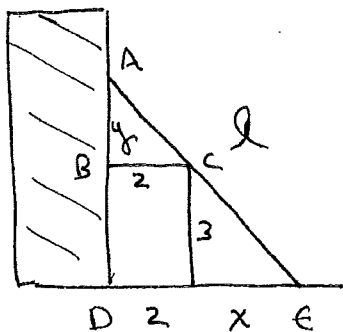
$$y^3 + 8y + 24 = 0$$

$$(y + 2)(y^2 - 2y + 12) = 0$$

$$y = -2 \Rightarrow x = -1$$

$$\therefore \text{pt @ } (-1, -2)$$

2.



$$\triangle ABC \sim \triangle ADE$$

$$\angle A = \angle A$$

$$\angle B = \angle D = 90^\circ$$

$$\frac{y}{y+3} = \frac{2}{x+2}$$

$$y(x+2) = 2(y+3)$$

$$yx + 2y = 2y + 6$$

$$x = \frac{6}{y}$$

$$l^2 = (y+3)^2 + (x+2)^2$$

$$l = \sqrt{(y+3)^2 + \left(\frac{6}{y} + 2\right)^2}$$

$$l' = \frac{2(y+3) - \frac{12}{y^2} \left(\frac{6}{y} + 2\right)}{2 \sqrt{(y+3)^2 + \left(\frac{6}{y} + 2\right)^2}}$$

$$2(y+3) - \frac{12}{y^2} \left(\frac{6}{y} + 2\right) = 0$$

$$y^4 + 3y^3 - 12y - 36 = 0$$

$$y \doteq 2.29 \text{ m}$$

$$\therefore l \doteq 7.02 \text{ m}$$

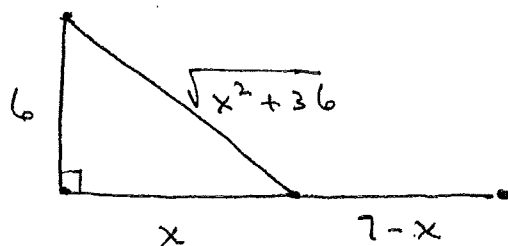
inadmissible $(y+3)(y^3-12)=0$

$$\rightarrow y = -3 ; y = \sqrt[3]{12}$$

BEST TIME

9

1.



$$T(x) = \frac{\sqrt{x^2 + 36}}{4} + \frac{7-x}{5}$$

$$T'(x) = \frac{x}{4\sqrt{x^2 + 36}} - \frac{1}{5}$$

$$\frac{x}{4\sqrt{x^2 + 36}} - \frac{1}{5} = 0$$

$$25x^2 = 16(x^2 + 36)$$

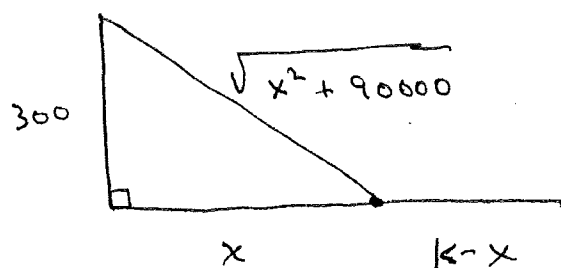
$$9x^2 = 576$$

$$x^2 = 64$$

$$x = 8 \text{ miles}$$

$\therefore$ He cannot reach the store in the least possible time since $x = 8 > 7$.

2.



$$T = \frac{\sqrt{x^2 + 90000}}{3} + \frac{K-x}{5}$$

$$T' = \frac{x}{3\sqrt{x^2 + 90000}} - \frac{1}{5}$$

$$\frac{x}{3\sqrt{x^2 + 90000}} - \frac{1}{5} = 0$$

$$25x^2 = 9x^2 + 810,000$$

$$16x^2 = 810,000$$

$$x = 225 \text{ m.}$$

$\therefore$ She should land 225m down the shore from the point on the shore directly across from her start point.

Profits, Cost, Revenue.

(11)

$$1. \quad p(x) = \frac{800,000 - x}{200,000}$$

$$C(x) = 125,000 + 0.42x$$

$$P(x) = R(x) - C(x)$$

$$P(x) = x \cdot p(x) - C(x)$$

$$P(x) = x \left(\frac{800,000 - x}{200,000} \right) - 125,000 - 0.42x$$

$$P(x) = 4x - \frac{x^2}{200,000} - 125,000 - 0.42x$$

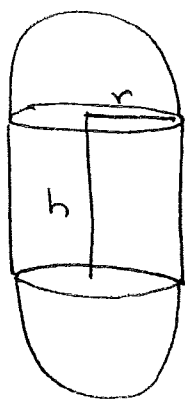
$$P'(x) = 4 - \frac{x}{100,000} - 0.42$$

$$\frac{x}{100,000} - 3.58 = 0$$

$$x = 358,000$$

$\therefore$ produce 358,000 widgets.

2.



$$V = \pi r^2 h + \frac{4}{3} \pi r^3 = 100$$

$$h = \frac{100 - \frac{4}{3} \pi r^3}{\pi r^2}$$

$$SA = 4\pi r^2 + 2\pi r h + 2\pi r^2$$

$$SA = 6\pi r^2 + 2\pi r \left(\frac{100 - \frac{4}{3} \pi r^3}{\pi r^2} \right)$$

$$SA = 6\pi r^2 + \frac{200}{r} - \frac{8}{3} \pi r^2$$

$$(SA)' = 12\pi r - \frac{200}{r^2} - \frac{16}{3} \pi r$$

$$\frac{20\pi r}{3} - \frac{200}{r^2} = 0$$

$$20\pi r^3 - 600 = 0$$

$$r^3 = \frac{30}{\pi}$$

$$r = \sqrt[3]{\frac{30}{\pi}}$$

$$r \approx 2.12 \text{ m}$$

$$h \approx 4.26 \text{ m}$$

3. $V = 1 \text{ Litre}$
 $V = 1000 \text{ cm}^3$ * $1 \text{ L} = 1000 \text{ cm}^3$

$$V = \pi r^2 h \rightarrow \text{cylindrical can.}$$

$$\pi r^2 h = 1000$$

$$h = \frac{1000}{\pi r^2}$$

$$SA = 2\pi r^2 + 2\pi r h$$

$$SA = 2\pi r^2 + 2\pi r \left(\frac{1000}{\pi r^2} \right)$$

$$SA = 2\pi r^2 + \frac{2000}{r}$$

$$(SA)' = 4\pi r - \frac{2000}{r^2}$$

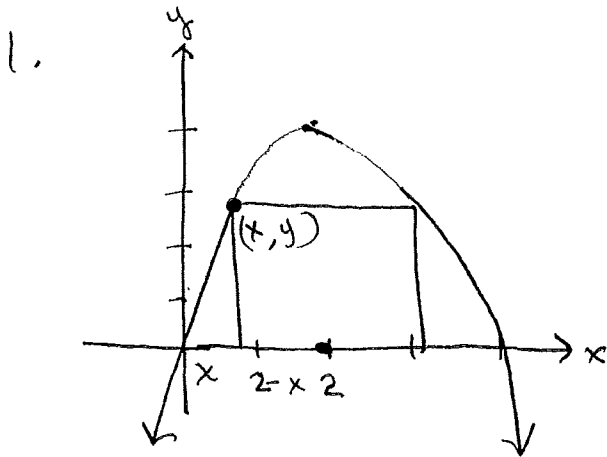
$$4\pi r - \frac{2000}{r^2} = 0$$

$$4\pi r^3 - 2000 = 0$$

$$r^3 = \frac{500}{\pi}$$

$$r = \sqrt[3]{\frac{500}{\pi}} \text{ cm.}$$

Additional Questions



$$y = 4x - x^2$$

$$\text{length} = 4 - 2x$$

$$\text{width} = y = 4x - x^2$$

$$A = L \cdot w$$

$$A = (4 - 2x)(4x - x^2)$$

$$A' = -2(4x - x^2) + (4 - 2x)(4 - 2x)$$

$$A' = -8x + 2x^2 + 16 - 16x + 4x^2$$

$$A' = 6x^2 - 24x + 16$$

$$A' = 3x^2 - 12x + 8$$

$$3x^2 - 12x + 8 = 0$$

$$x = \frac{6 \pm 2\sqrt{3}}{3}$$

$$x \doteq 3.15 \text{ or } x \doteq 0.85$$

↑
inadmissible

$$\therefore \text{Length} = 4 - 2(0.85)$$

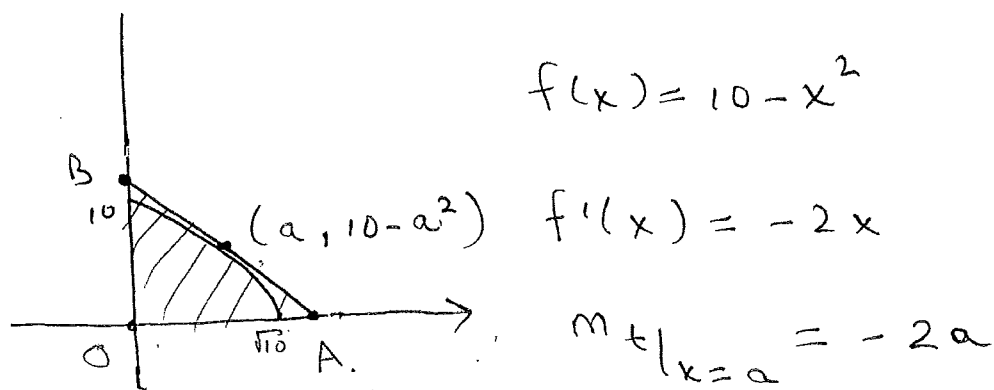
$$\boxed{\text{Length} \doteq 2.31 \text{ units}}$$

$$\text{width} = 4(0.85) - (0.85)^2$$

$$\boxed{\text{width} \doteq 2.68 \text{ units}}$$

(2)

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Equation of Tangent line at $(a, f(a))$ is

$$y - (10 - a^2) = -2a(x - a)$$

* x-int: set $y = 0$

$$-(10 - a^2) = -2ax + 2a^2$$

$$x = \frac{a^2 + 10}{2a} \quad \therefore A\left(\frac{a^2 + 10}{2a}, 0\right)$$

* y-int: set $x = 0$

$$y - (10 - a^2) = 2a^2$$

$$y = a^2 + 10 \quad \therefore B(0, a^2 + 10)$$

$$A = \frac{1}{2} b \cdot h$$

$$A = \frac{1}{2} \left(\frac{a^2 + 10}{2a} \right) (a^2 + 10)$$

$$A = \frac{(a^2 + 10)^2}{4a}$$

$$A' = \frac{1}{4} \left[\frac{4a(a^2 + 10)(a) - (a^2 + 10)^2}{a^2} \right]$$

over

②
cont'd

$$A' = 0$$

16

$$(a^2 + 10)(4a^2 - (a^2 + 10)) = 0$$

$$(a^2 + 10)(3a^2 - 10) = 0$$

$$3a^2 - 10 = 0$$

$$a = \sqrt{\frac{10}{3}} \text{ or } -\sqrt{\frac{10}{3}}$$

↑ inadmissible

∴ Equation is:

$$y - \frac{20}{3} = -2\sqrt{\frac{10}{3}} \left(x - \sqrt{\frac{10}{3}} \right)$$

(OR)

$$y - \frac{20}{3} = -\frac{2\sqrt{30}}{3} \left(x - \frac{\sqrt{30}}{3} \right)$$